FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
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ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2015**

# HALF COURSE TEST –VIII

## Paper 1

**Time Allotted: 3 Hours** **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

## INSTRUCTIONS

### A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-B**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

### B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

### C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   **Section – A (09 – 12)** contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

   **Section – A (13 – 18)** contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 marks** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 mark** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.

| | |
|---|---|
| **Name of the Candidate** | |
| **Enrolment No.** | |

## Useful Data

### PHYSICS

| | |
|---|---|
| Acceleration due to gravity | $g = 10\ m/s^2$ |
| Planck constant | $h = 6.6 \times 10^{-34}$ J-s |
| Charge of electron | $e = 1.6 \times 10^{-19}$ C |
| Mass of electron | $m_e = 9.1 \times 10^{-31}$ kg |
| Permittivity of free space | $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$ |
| Density of water | $\rho_{water} = 10^3\ kg/m^3$ |
| Atmospheric pressure | $P_a = 10^5\ N/m^2$ |
| Gas constant | $R = 8.314\ J\ K^{-1}\ mol^{-1}$ |

### CHEMISTRY

| | | | |
|---|---|---|---|
| Gas Constant | R | = | $8.314\ J\ K^{-1}\ mol^{-1}$ |
| | | = | $0.0821$ Lit atm $K^{-1}\ mol^{-1}$ |
| | | = | $1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$ |
| Avogadro's Number | $N_a$ | = | $6.023 \times 10^{23}$ |
| Planck's constant | h | = | $6.625 \times 10^{-34}$ J·s |
| | | = | $6.625 \times 10^{-27}$ erg·s |
| 1 Faraday | | = | 96500 coulomb |
| 1 calorie | | = | 4.2 joule |
| 1 amu | | = | $1.66 \times 10^{-27}$ kg |
| 1 eV | | = | $1.6 \times 10^{-19}$ J |

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

# Physics

## PART – I

### SECTION - A

**Straight Objective Type**

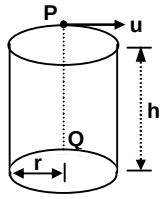
*This section contains 8 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

1. The equation of a particle executing SHM is given by $x = 3\cos(\pi/2)t$ cm, here t is in second. The distance travelled by the particle in first 8.5 sec is

(A) $24 + \frac{3}{\sqrt{2}}$cm (B) $27 - \frac{3}{\sqrt{2}}$ cm

(C) $24 - \frac{3}{\sqrt{2}}$ cm (D) $27 + \frac{3}{\sqrt{2}}$cm

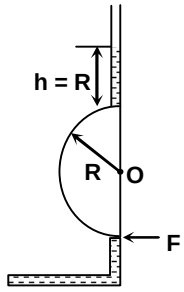
2. A hollow vertical cylinder of radius r and height h has smooth internal surface. A small particle is placed in contact with the inner side of the upper rim, at a point P. It is given a horizontal speed u, tangential to rim. It leaves the lower rim at point Q, vertically below P. the number of revolutions made by the particle will be




(A) $n = \frac{h}{2\pi r}$ (B) $n = \frac{u}{\sqrt{2gh}}$

(C) $n = \frac{2\pi r}{h}$ (D) $n = \frac{u}{2\pi r}\sqrt{\left(\frac{2h}{g}\right)}$

3. The figure shows a semi-cylindrical gate pivoted at the point O holding a stationary liquid of density $\rho$. A horizontal force F is applied at its lowest position to kept it stationary. The magnitude of the force is

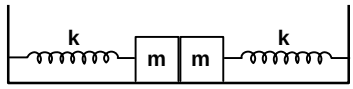



(A) $\frac{9}{2}\rho gR^2$ (B) $\frac{3}{2}\rho gR^2$

(C) $\rho gR^2$ (D) zero

***Space for rough work***

4. An assembly of identical spring mass system is placed on a smooth horizontal surface as shown. Initially the springs are relaxed. The left mass is displaced to the left while the right mass is displaced to right by two times more than the left spring and both are released. The resulting collision is elastic. The time period of the oscillation of the system is




(A) $2\pi\sqrt{\frac{2M}{k}}$ (B) $2\pi\sqrt{\frac{M}{2k}}$

(C) $2\pi\sqrt{\frac{M}{k}}$ (D) $\pi\sqrt{\frac{M}{k}}$

5. A body of mass m is free to move in a conservative field with potential energy $U = U_0(1 - \cos \alpha x)$ where $U_0$ and $\alpha$ are constants the time period of small oscillations is

(A) $2\pi\sqrt{\frac{U_0}{m\alpha^2}}$ (B) $2\pi\sqrt{\frac{m\alpha^2}{U_0}}$

(C) $2\pi\sqrt{\frac{m}{U_0\alpha^2}}$ (D) $2\pi\sqrt{\frac{U_0\alpha^2}{m}}$

6. A metallic sphere of mass 2 gm falls through a liquid with terminal velocity v. If we drop a sphere of mass 0.25 gm of the same metal through same liquid its terminal velocity would be

(A) v (B) 2v

(C) v/2 (D) v/4

7. An artificial satellite of mass m is moving in circular orbit at a height equal to the radius R of the earth. Suddenly due to internal explosion the satellite breaks into two parts of equal masses. One part of the satellite stops just after the explosion and then falls to the surface of the earth. The increase in the mechanical energy of the system (satellite + earth) due to explosion will be (g is the acceleration due to gravity on the surface of earth)

(A) mgR (B) $\frac{mgR}{2}$

(C) $\frac{mgR}{4}$ (D) $\frac{3mgR}{4}$

***Space for rough work***

8. A particle of mass equal to half of the earth's mass is released at a distance h(h □ R) from the surface of earth. The time after which the particle will hit earth's surface is

(A) $\sqrt{\frac{4h}{3g}}$ (B) $\sqrt{\frac{2h}{g}}$

(C) $\sqrt{\frac{h}{g}}$ (D) $\sqrt{\frac{2h}{3g}}$

**(Multiple Correct Choice Type)**

This section contains ***4 multiple choice questions***. Each question has four choices (A), (B), (C) and (D) out of which ***ONE or MORE*** are correct.

9. A satellite is revolving around the earth in a circular orbit. The universal gravitational constant suddenly becomes zero at time t = 0. Then
(A) The kinetic energy for t < 0 and t > 0 is same
(B) The angular momentum for t < 0 and t > 0 is same
(C) The angular momentum keeps changing for t < 0
(D) The angular momentum keeps changing for t > 0

10. Two waves of same frequency and wavelength but amplitude $A_1$ and $A_2$ meet a point at the same time. According to the superposition principle, the resultant amplitude
(A) may be $A_1 + A_2$ (B) may be $A_1 - A_2$
(C) $A_1 + A_2$ or $A_1 - A_2$ (D) $A_1 - A_2 \le A \le A_1 + A_2$

11. When tuning forks A and B are struck, a beat frequency $f_{AB}$ is heard. When A and C are struck a beat frequency of $f_{AC}$ is heard. It was found that $f_{AC} > f_{AB}$. If $f_A$, $f_B$ and $f_C$ are individual frequencies, then what are the possible relations?
(A) $f_A > f_B > f_C$ (B) $f_A < f_B < f_C$
(C) $f_B > f_A > f_C$ (D) $f_C > f_A > f_B$

12. A body of mass M is attached to the lower end of a metal wire, whose upper end is fixed. The mass M is brought slowly to the equilibrium position and the elongation in the wire is $\ell$.
(A) the energy stored in the wire is $Mg\ell$ (B) the energy stored in the wire is $\frac{1}{2}Mg\ell$
(C) heat generated during the process is $\frac{1}{2}Mg\ell$ (D) heat generated during the process is $\frac{1}{4}Mg\ell$

***Space for rough work***

## Linked Comprehension Type

*This section contains **2 paragraphs**. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

### Paragraph for Question Nos. 13 to 15

If a body moves through a liquid or a gas then the fluid applies a force on the body which is called drag force. Direction of the drag force is always opposite to the motion of the body relative to the fluid. At low speeds of the body, drag force ($F_P$) is directly proportional to the speed.

$F_D = kv$

When k is a proportionally constant and it depends upon the dimension of the body moving in air at relatively high speeds, the drag force applied by air on the body is proportional to $v^2$ instead of v.

$F_D = k_1v^2$

Where this proportionally constant K can be given by

$k_1 = \rho CA$

Where $\rho$ is the density of air
C is another constant giving the drag property of air
A is area of cross-section of the body

Consider a case an object of mass m is released from a height h and it falls under gravity. As it's speed increases the drag force starts increasing on the object. Due to this at some instant, the object attains equilibrium. The speed attained by the body at this instant is called "terminal speed" of the body.
Assume that the drag force applied by air on the body follows the relation $F_D = kv$, neglect the force of buoyancy applied by air on the body then answer the following questions.

***Read the above passage carefully and answer the following questions***

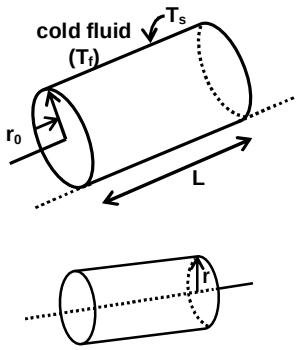
13. What is the pattern of acceleration change of the body?
(A) It first increases, then decreases to zero
(B) It decreases uniformly from starting to zero
(C) It decrease non-uniformly from starting to zero
(D) It first increases then becomes constant

14. What is the terminal speed of the object?
(A) $\frac{mg}{k}$ (B) mgk
(C) $\frac{k}{mg}$ (D) $\sqrt{\frac{mg}{k}}$

15. What is the nature of change in speed of the body?
(A) It first decreases and then increases to a constant value
(B) It increases uniformly to a constant value
(C) It increases non-uniformly to a constant value
(D) It continuously decreases to zero.

***Space for rough work***

**Paragraph for Question Nos. 16 to 18**




Heat generation may occur in a variety of radial geometries. Consider a long, solid cylinder as shown in the figure, which could represent a current-carrying wire or a fuel element in a nuclear reactor. For steady state conditions, the rate at which heat is generated within the cylinder must equal the rate at which heat is convected from the surface of the cylinder to a moving fluid. This condition allows the surface temperature to be maintained at a fixed value of $T_S$.

To determine the temperature distribution in the cylinder, we begin with energy conservation principle. Consider a cylindrical section of radius r. The rate at which energy is generated within the given volume is conducted radially outward.

$\therefore \quad \dot{q}\,\pi\, r^2\ell = -K2\pi r\ell \frac{dT}{dr}$

where $\dot{q}$ is the energy generated per unit time per unit volume, K is the thermal conductivity and dT/dr is the temperature gradient at radius r.

If $\dot{q}$ is constant

$$T(r) = -\frac{\dot{q}}{4k}r^2 + C$$

At $r = r_0, \quad T(r_0) = T_S$

Therefore, $T(r) = \frac{\dot{q}}{4k}r_0^2\left(1-\frac{r^2}{r_0^2}\right) + T_S$

The rate of heat convected to the surrounding fluid (at temperature $T_f$) by the surface at temperature $T_s$ is proportional to the temperature difference $(T_S - T_f)$ and the surface area in contact with the fluid.

Thus, rate of heat convection = $h\,(2\pi\, r_0 L)\,(T_S - T_f)$

where h is a constant called heat convection coefficient.

By overall energy balance,

$\dot{q}\left(\pi r_0^2 L\right) = h\left(2\pi r_0\ell\right)\left(T_S - T_f\right)$

$\Rightarrow T_S = T_f + \frac{\dot{q}r_0}{2h}$

**Read the above passage carefully and answer the following questions.**

16. Consider a spherical nuclear fuel material of radius $r_0$ and rate of heat generation $\dot{q}$ (watt/$m^3$). Find the temperature difference between the surface and surrounding fluid in steady state (h is the heat convection coefficient).

(A) $\frac{\dot{q}r_0}{2h}$ (B) $\frac{2}{3}\frac{\dot{q}r_0}{h}$

(C) $\frac{\dot{q}r_0}{3h}$ (D) $\frac{\dot{q}r_0}{h}$

***Space for rough work***

17. In the above passage, the ratio of temperature gradient at $r = r_0/2$ and $r = r_0$ is

(A) 1 (B) 1/4

(C) 1/2 (D) 1/8

18. In the given passage, the difference in temperature at the axis and surface of the cylinder is

(A) $\frac{\dot{q}r_0^2}{4k}$ (B) $\frac{\dot{q}r_0^2}{k}$

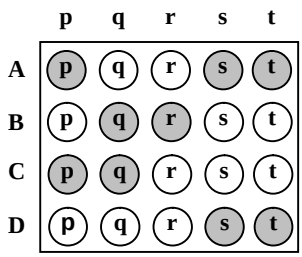
(C) $\frac{\dot{q}r_0^2}{2k}$ (D) $\frac{2\dot{q}r_0}{k}$

***Space for rough work***

## SECTION - B

### Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:




If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Consider a situation (i) that two sound waves, $y_1 = (0.2m) \sin 504\pi\left(t - \frac{x}{300}\right)$ and $y_2 = (0.6\ m) \sin 496\pi\left(t - \frac{x}{300}\right)$ are super imposed. Consider another situation (ii) that two sound waves $y_1 = (0.4m) \sin 504\pi\left(t - \frac{x}{300}\right)$ and $y_2 = (0.4\ m) \sin 504\pi\left(t + \frac{x}{300}\right)$ are superimposed. Match the following.

| **Column – I** | **Column – II** |
|---|---|
| (A) In situation (i) | (p) Stationary waves are formed. |
| (B) In situation (ii) | (q) There will be phenomenon of beats |
| (C) When two waves of same frequency and amplitude and travelling in opposite directions superimpose. | (r) Amplitude of the resultant wave will vary periodically with position |
| (D) If the intensity of sound alternately increases and decreases periodically a result of superposition of waves of highly different frequencies. | (s) Amplitude of the resultant wave will vary periodically with time |
| | (t) amplitude of the resultant wave will be constant |

***Space for rough work***

2. Match the following:

| Column I | Column II |
|---|---|
| (A) A cone resting with its slant surface touching against the smooth horizontal floor | (p) Stable Equilibrium |
| (B) A car moving on a banked smooth circular track of banking angle $\theta$ with constant speed without its tyres slipping | (q) Unstable equilibrium |
| (C) A rigid body suspended in a laboratory in such a manner that its centre of gravity lies vertically above point of suspension and its centre of mass lies vertically below point of suspension. | (r) Neutral equilibrium |
| (D) A uniform cube balanced on a horizontal floor on one of its edge (only a force parallel to horizontal plane can be applied). | (s) none of the above |
| | (t) potential energy of the system remains constant. |

***Space for rough work***

# Chemistry

## PART – II

## SECTION - A

### Straight Objective Type

*This section contains 8 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

1. At 1400 K, $K_c = 2.5 \times 10^{-3}$ for the following reaction:

$$CH_4(g) + 2H_2S(g) \rightleftharpoons CS_2(g) + 4H_2(g)$$

If a 10 litre reaction vessel at 1400 K contains 2.0 moles of $CH_4$, 3.0 moles of $CS_2$, 3.0 moles of $H_2$ and 4.0 moles of $H_2S$, then equilibrium concentration of $CS_2$ will be

(A) = 0.3 M (B) < 0.3 M
(C) > 0.3 M (D) none of these

2. If the dipole moment of 1, 2, 3, 5 – tetrachloro benzene is xD, then the dipole moment of chlorobenzene will be

(A) 0.5 xD (B) 0.25 xD
(C) xD (D) 1.25 xD

3. A sample of four hydrogen atoms, all in the nth excited state, de – excite to the ground state, and shows a maximum of six different spectral lines. Hence the value of 'n' is

(A) 3 (B) 4
(C) 5 (D) 6

4. A mixture containing ferrous oxalate & ferrous sulphate required 100 ml of 0.1M acidified $KMnO_4$ solution. This mixture is now reduced by $SnCl_2$ so that all $Fe^{3+}$ changes to $Fe^{2+}$ ions & the resultant mixture now required 50 ml of the same permanganate solution, the percentage by mass of ferrous oxalate in the mixture is (Fe = 56, C = 12, O = 16, S = 32)

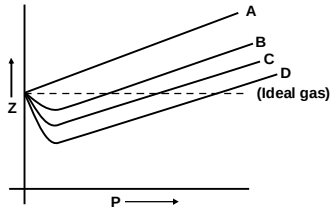
(A) 18.6% (B) 28.6%
(C) 38.6% (D) 48.6%

***Space for rough work***

5. Identify the incorrect statement among the following:
   (A) If Auf- bau rule is not followed, then Sn(atomic number = 50) would be placed in d – block of the periodic table
   (B) $5^{th}$ ionization energy of As (Z = 33) is less than that of Ge (Z = 32)
   (C) Increasing order of first I.E. of S, Se & Cl is Se < S < Cl
   (D) Ionic radii of $Mg^{2+}$ (Z = 12) is smaller than $F^-$ (Z = 9)

6. 150 ml of 0.1 M NaOH solution is mixed with 100 ml of 0.1 M $H_2CO_3$ solution, if $K_{a_1}$ and $K_{a_2}$ of $H_2CO_3$ are $10^{-7}$ and $10^{-13}$ respectively, then pOH of the resulting solution will be
   (A) 13 (B) 7
   (C) 1 (D) 2

7. Compressibility factor (Z) is plotted against pressure (p) for four different gases A, B, C & D. The correct order of critical temperature of the gases shown in the above graph is
   (A) A > B > C > D
   (B) B < A < C < D
   (C) A < B < C < D
   (D) D < B < C < A




8. Which of the following compounds exhibits tautomerism?
   (A) chloroethane (B) ethanol
   (C) ethoxyethane (D) nitroethane

***Space for rough work***

**(Multiple Correct Choice Type)**

This section contains ***4 multiple choice questions***. Each question has four choices (A), (B), (C) and (D) out of which ***ONE or MORE*** are correct.

9. Which of the following mixture is/are buffers?
   (A) 100 ml of 0.1 M $Ca(CH_3COO)_2$ & 100 ml of 0.1 M HCl
   (B) 100 ml of 0.1 M $(PhNH_3)_2SO_4$ & 50 ml 0.1 M $Ca(OH)_2$
   (C) 100 ml of 0.1 M $(NH_4)_2SO_4$ & 100 ml of 0.1 M $NH_4OH$
   (D) 100 ml of 0.1 M borax solution ($Na_2B_4O_7.10H_2O$) & 25 ml of 0.05 M $H_2SO_4$

10. Which of the following equilibria shifts in the forward direction on decreasing the pressure?
   (A) $\text{ice} \rightleftharpoons \text{water}$
   (B) $\text{water} \rightleftharpoons \text{water vapour}$
   (C) $S(\ell) \rightleftharpoons S(s)$
   (D) $\text{diamond} \rightleftharpoons \text{graphite}$ (density of diamond is more than graphite)

11. The percentage by mass of chlorine in the chloride of a metal is 80%. 33.375 g of this chloride on vaporization occupies a volume of 5.6 litre at NTP. Now select the correct options given below:
   (A) valency of the metal is 2
   (B) molecular weight of the metal chloride is 133.5
   (C) atomic weight of the metal is 27 g
   (D) 9 g of the metal can completely react with 8 g of oxygen

12. Identify the correct statement (s) among the following:
   (A) The electronegativity of an element is the tendency of an isolated atom to attract an electron
   (B) The second ionization potential of boron is greater than carbon
   (C) Nitrogen has almost zero electron gain enthalpy
   (D) The first electron affinity of fluorine is greater than that of chlorine

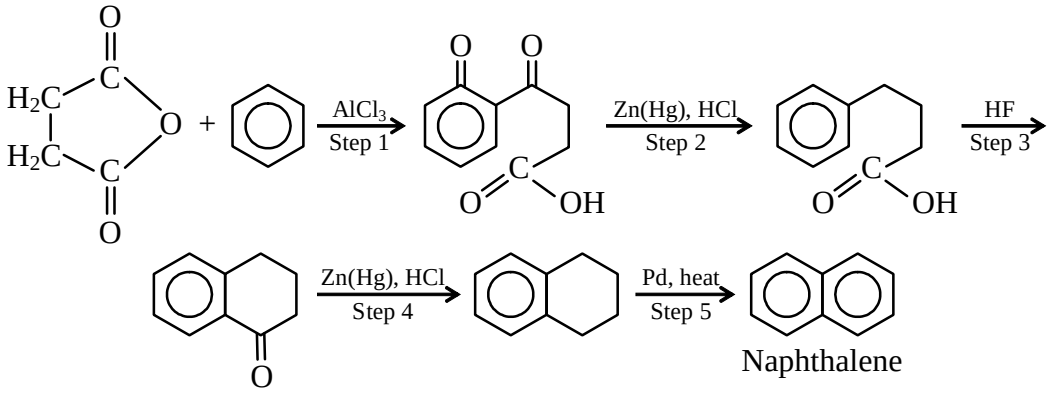
***Space for rough work***

## Linked Comprehension Type

*This section contains **2 paragraphs**. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

### Paragraph for Question Nos. 13 to 15

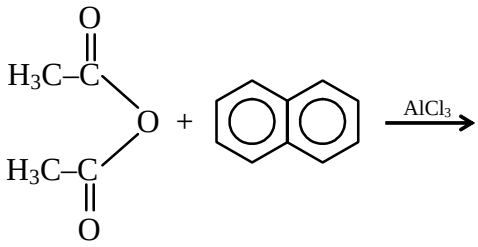
The Lewis acid usually reacts with the acyl halide to form an acylium ion. The acylium ion is stabilized by resonance. The acylium ion acts as an electrophile attacking the benzene ring to form an arenium ion. The arenium ion then loses a proton to generate the final product. Powerful electron withdrawing groups on the benzene ring such as another acyl group will retard this reaction. Naphthalene is the simplest and most important of the fused ring hydrocarbons. Five percent of all constituents of coal tar are naphthalene. Naphthalene can be manufactured using the Friedel-Crafts reaction via the reaction pathway shown below:

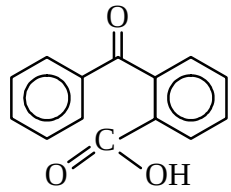


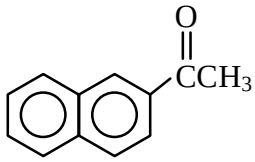

13. The Friedel-Crafts reaction occurs twice in the given sequence of reactions. Which two steps represent Friedel-Crafts reactions?
    (A) Steps 1 and 3 (B) Steps 1 and 5
    (C) Steps 2 and 4 (D) Steps 3 and 5

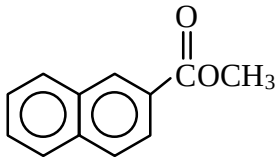
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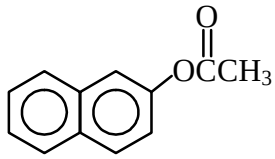
14. What are the most likely products of the following reaction?




(A) $C_2H_6$ + 


(B) $CH_3COH$ + 


(C) $CH_3CH$ + 


(D) $CH_3CH$ + 


15. Which of the following is an acylium ion?

(A) $AlCl_4^-$

(B) 

(C) 

(D) 

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**Paragraph for Question Nos. 16 to 18**

The solubility of some sparingly soluble salts e.g. AgCl, AgBr, AgI, $Ag_2CrO_4$ etc can be appreciably increased through complexation reaction. Let's take the case of AgCl which is very feebly soluble in water ($K_{sp}$ of AgCl = $1 \times 10^{-10}$ at 25°C) but if liquid ammonia is added into this solution, its solubility increased so many folds due to the formation of complex $\left[Ag(NH_3)_2\right]^+$.

$$AgCl + 2NH_3(aq) \rightleftharpoons \left[Ag(NH_3)_2\right]^+ + Cl^-$$

A solution of $Ag^+$ ions at a concentration of $4.0 \times 10^{-13}$ M just fails to yield a precipitate of AgCl with concentration of $1.0 \times 10^{-3}$ M of $Cl^-$ ions, when the concentration of ammonia in the solution is $2.0 \times 10^{-2}$ M.

**Now, answer the following questions.**

16. The solubility of AgCl in water **(**without adding $NH_3$) at 25°C is
(A) $10^{-4}$ M (B) $10^{-5}$ M
(C) $10^{-6}$ M (D) $10^{-8}$ M

17. The dissociation constant of the complex $\left[Ag(NH_3)_2\right]^+$ at 25°C is
(A) $10^{-7}$ (B) $10^{+7}$
(C) $10^{-8}$ (D) $10^{+8}$

18. $Ag^+ + 2NH_3 \rightleftharpoons \left[Ag(NH_3)_2\right]^+$

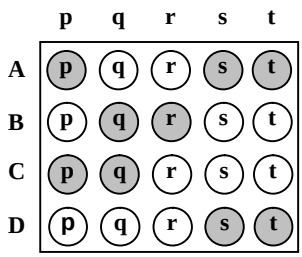
On adding more $NH_3$, the above equilibrium shifts:
(A) In the forward direction (B) In the backward direction
(C) The complex dissociation faster (D) None of these

***Space for rough work***

## SECTION - B

### Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:




If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the statement in column – I with corresponding type (s) of silicates in column - II

| Column – I | | Column – II | |
|---|---|---|---|
| (A) | Only one corner of $SiO_4^{4-}$ units is shared | (p) | Chain silicates |
| (B) | At least one corner of $SiO_4^{4-}$ unit is shared | (q) | Sheet silicates |
| (C) | Only two corners of $SiO_4^{4-}$ units are shared | (r) | Pyrosilicates |
| (D) | At least two corners of $SiO_4^{4-}$ units are shared | (s) | Cyclic silicates |

2. Match the compounds in column – I with their corresponding decreasing $K_a$ value in column – II.

| Column – I (compounds) | | Column – II (Acidity) | |
|---|---|---|---|
| (A) | o, m, p – nitrophenols | (p) | o > m > p |
| (B) | o, m, p – cresols | (q) | p > o > m |
| (C) | o, m, p – chlorophenols | (r) | p > m > o |
| (D) | o, m, p – nitrobenzoic acid | (s) | m > p > o |

***Space for rough work***

# Mathematics

## PART – III

## SECTION - A

### Straight Objective Type

*This section contains 8 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

1. If the roots of $ax^2 - bx - c = 0$ change by the same quantity then the expression in a, b, c that does not change is

(A) $\frac{b^2 - 4ac}{a^2}$ (B) $\frac{b - 4c}{a}$

(C) $\frac{b^2 + 4ac}{a^2}$ (D) $\frac{a^2 + b^2}{c^2}$

2. The point of intersection of the curves $\arg(z - 3i) = \frac{3\pi}{4}$ and $\arg(2z + 1 - 2i) = \frac{\pi}{4}$ is

(A) $\frac{1}{4}(3 + 9i)$ (B) $\frac{1}{4}(3 - 9i)$

(C) $\frac{1}{2}(3 + 2i)$ (D) none of these

3. Suppose a, b, c are in A.P. and $a^2$, $b^2$, $c^2$ are in G.P. If $a < b < c$ and $a + b + c = \frac{3}{2}$, then the value of a is

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2\sqrt{3}}$

(C) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (D) $\frac{1}{2} - \frac{1}{\sqrt{2}}$

***Space for rough work***

4. If {x} represents fractional part of x, then $\left\{\frac{5^{200}}{8}\right\}$ is

(A) $\frac{1}{4}$ (B) $\frac{1}{8}$

(C) $\frac{3}{8}$ (D) $\frac{5}{8}$

5. The maximum number of points of intersection of 8 circles is

(A) 16 (B) 24

(C) 28 (D) 56

6. If $a \sin x + b \cos(x + \theta) + b \cos(x - \theta) = d$ for some real x, then the minimum value of $|\cos \theta|$ is

(A) $\frac{1}{2|b|}\sqrt{d^2 - a^2}$ (B) $\frac{1}{2|a|}\sqrt{d^2 - a^2}$

(C) $\frac{1}{2|d|}\sqrt{d^2 - a^2}$ (D) $\frac{1}{|a|}\sqrt{d^2 - a^2}$

7. The number of real solutions of the equation $\tan^{-1}\sqrt{x^2 - 3x + 2} + \sin^{-1}\sqrt{4x - x^2 - 3} = \pi$ is

(A) one (B) two

(C) zero (D) infinite

8. If in a triangle $a^2 + b^2 + c^2 = ac + \sqrt{3}ab$, then the triangle is

(A) equilateral (B) right angled isosceles

(C) right angled scalene (D) none of these

***Space for rough work***

**(Multiple Correct Choice Type)**

This section contains ***4 multiple choice questions***. Each question has four choices (A), (B), (C) and (D) out of which ***ONE or MORE*** are correct.

9. The roots of $ax^2 + bx + c = 0$, are non–real complex and $a + c < b$, then
(A) $4a + c > 2b$ (B) $4a + c < 2b$
(C) $a + 4c > 2b$ (D) $a + 4c < 2b$

10. A circle of radius r ($\neq 0$) touches the parabola $y^2 + 12x = 0$ at the vertex of the parabola. The centre of the circle lies to the left of the vertex and the circle lies entirely within the parabola, then the interval(s) in which r lies can be
(A) (1, 7) (B) (0, 6)
(C) $\left(1, \frac{13}{2}\right)$ (D) (2, 8)

11. If $\left|\frac{2iz_1 - z_1 - z_2}{2iz_1 + z_1 + z_2}\right| = \left|\frac{\cos\theta + i\sin\theta}{\cos\theta - i\sin\theta}\right|$ and $z_1$, $z_2$ are non-zero complex numbers then
(A) $\frac{z_1 + z_2}{z_1}$ is purely real (B) $\frac{z_1}{z_2}$ is purely real
(C) $\frac{z_1 + z_2}{z_1 - z_2}$ is purely real (D) none of these

12. The equation $\left|\sqrt{x^2 + (y-1)^2} - \sqrt{x^2 + (y+1)^2}\right| = k$ will represent a hyperbola for
(A) $k \in (0, 2)$ (B) $k \in (0, 1)$
(C) $k \in (1, \infty)$ (D) $k \in (0, \infty)$

***Space for rough work***

## Linked Comprehension Type

*This section contains **2 paragraphs**. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

### Paragraph for Question Nos. 13 to 15

**Read the following write up carefully and answer the following questions:**

Let $\bar{r}$ be the position vector of a variable point such that $\bar{r} \cdot (10j - 8i - \bar{r}) = 40$ and $P_1 = \max\{|\bar{r} + 2i - 3j|\}$, $P_2 = \min\{|\bar{r} + 2i - 3j|^2\}$. A tangent is drawn to the curve $y = \frac{8}{x^2}$ at the point A with abscissa 2. The drawn line cuts the x-axis at B

13. The value of $P_2$ is equal to
    (A) 9 (B) $2\sqrt{2}-1$
    (C) $6\sqrt{2}+3$ (D) $9-4\sqrt{2}$

14. The value of $P_1 + P_2$ is equal to
    (A) 2 (B) 10
    (C) 18 (D) 5

15. $\overrightarrow{AB} \cdot \overrightarrow{OB}$ is equal to
    (A) 1 (B) 2
    (C) 3 (D) 4

***Space for rough work***

**Paragraph for Question Nos. 16 to 18**

**Read the following write up carefully and answer the following questions:**
Consider the complex numbers $z_1$ and $z_2$ satisfying the relation $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$

16. Complex number $z_1\bar{z}_2$ is
(A) purely real
(B) purely imaginary
(C) zero
(D) none of these

17. $\frac{z_1}{z_2}$ is
(A) purely real
(B) purely imaginary
(C) zero
(D) none of these

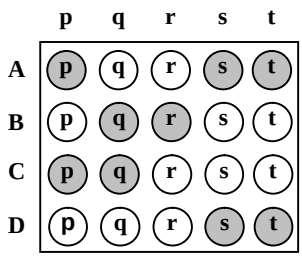
18. One of the possible argument of $i\left(\frac{z_1}{z_2}\right)$ is
(A) $\frac{\pi}{2}$
(B) $-\frac{\pi}{2}$
(C) 0
(D) none of these

***Space for rough work***

## SECTION - B

### Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:




If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. Match the following Column–I with Column–II

| Column – I | Column – II |
|---|---|
| (A) The locus of mid-points of chords of an ellipse which are drawn through an end of minor axis, is | (p) pair of straight line |
| (B) The locus of an end of latus rectum of all ellipse having a given major axis is | (q) circle |
| (C) The locus of the foot of perpendicular from a focus of the ellipse on any tangent is | (r) parabola |
| (D) A variable line drawn through a fixed point cuts axes at A and B. The locus of mid-point of AB is | (s) ellipse |
| | (t) hyperbola |

2. Match the following Column–I with Column–II

| Column – I | Column – II |
|---|---|
| (A) If the equation $x^2 + 4 + 3\sin(ax + b) - 2x = 0$ has atleast one real solution, where $a, b \in [0, 2\pi]$, then $\sin(a + b)$ can be equal to | (p) –1 |
| (B) $\sin^{-1} x \le \cos^{-1} x$, then x can be equal to | (q) 0 |
| (C) The number of the ordered pairs (x, y) satisfying $\lvert y\rvert = \cos x$ and $y = \sin^{-1}(\sin x)$, where $-2\pi \le x \le 3\pi$, is equal to | (r) 1 |
| (D) If $n \in N$ and the set of equations $\cos^{-1} x + (\sin^{-1} y)^2 = \frac{n\pi^2}{4}$ and $\left(\sin^{-1} y\right)^2 - \cos^{-1} x = \frac{\pi^2}{16}$ is consistent, then n can be equal to | (s) 4 |
| | (t) 5 |

***Space for rough work***